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Diffusion of Bottlebrush Polymer-Grafted Nanoparticles: The Role of Matrix Molecular Weight

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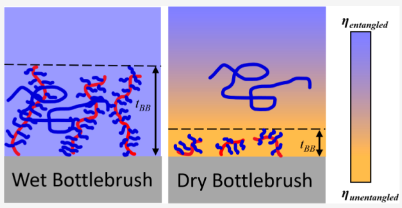
ABSTRACT: Nanoparticle diffusion in a polymer matrix is known to be sensitive to the surface chemistry and size of the nanoparticles, and this study explores the effect of grafted polymers on the nanoparticles. We report the diffusion coefficients of nanoparticles grafted with bottlebrush polymers in melts of linear polymers. These bottlebrush polymer-grafted nanoparticles (PGNs) are comprised of large silica NPs (radius 80 nm) grafted with various backbone degrees of polymerization and fixed polystyrene side chains (4.2 kDa) to give bottlebrush molecular weights of 80−388 kDa. The polymer matrix is polystyrene (100 or 425 kDa), and nanoparticle diffusion was studied at 170 °C. After annealing a trilayer sample, time-of-flight secondary ion mass spectrometry (ToF-SIMS) measured the cross-sectional PGN concentration profiles and extracted diffusion coefficients. The PGNs with high bottlebrush molecular weights relative to the matrix polymer display core−shell diffusion behavior, wherein the shell thickness is comparable to the grafted bottlebrush thickness. Surprisingly, when the bottlebrush molecular weight is smaller than the matrix polymer, the PGNs diffuse 10−100 times faster than predicted, which we attribute to a local decrease in viscosity associated with disentanglement of the matrix polymer near the large nanoparticles.



■ INTRODUCTION

Polymer nanocomposites, comprised of a polymer matrix with a nanoscale filler, are well-established materials due to their tunable properties, beginning with carbon black reinforced elastomers and continuing to expand with the introduction of new nanoparticles.[1] The mechanical, optical, and electronic properties in polymer nanocomposites (PNCs) are controlled by the nanoparticle properties, polymer matrix properties, and the spatial arrangement of the nanoparticles.[2,3] To support the emerging applications of PNCs, including self-healing materials, drug delivery systems, and electrochemical devices, the fundamental understanding of nanoparticle−polymer interactions must continue to develop.[1,4−6] Nanoparticle (NP) diffusion is of particular interest due to the important role it plays during PNC processing and extended use, and the connection to NP dispersion.

Attractive interactions between NPs and polymers are an effective strategy to circumvent NP aggregation and promote NP dispersion, because the strongly adsorbed polymer prevents NP-NP contact. Similarly, polymer-grafted nanoparticles (PGNs) promote nanoparticle dispersion, affecting the mechanical,[7−9] optical,[10] electrical,[11,12] and magnetic[13] properties of the resulting PNC. Linear PGNs and their role in structure−property relations in the melt state have been studied,[14−16] including their impact on polymer diffusion.[17,18] Nonlinear polymers have also been grafted to NPs, including bottlebrush polymers.[19−21] In addition to the grafting density and the total molar mass of grafted linear polymers, bottlebrush PGNs have additional tunability related to the backbone (red) and side chain (blue) degrees of polymerization (Figure 1). The versatility of bottlebrushes allows them

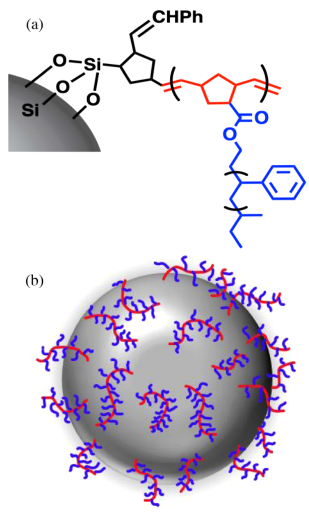


Figure 1. (a) Chemistry of bottlebrush polymers, with polynorbornene backbones (red) and polystyrene side chains (blue). (b) Schematic of bottlebrush polymer grafted nanoparticles (PGNs).

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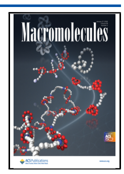

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to be explored as tissue-mimicking elastomers,[22,23] photonic crystals,[24,25] batteries,[26] triboelectric nanogenerators,[27] and dielectric elastomer actuators.[28] While important for processing and properties of PNCs, the diffusion of bottlebrush PGNs has not been studied previously. Moreover, the diffusion coefficients of bottlebrush PGNs could elucidate the nature of the bottlebrush in a polymer matrix.

In ideal systems of isolated particles, the diffusion of monodisperse particles in a fluid follows the hydrodynamic Stokes–Einstein (SE) equation:

$$D_{SE} = \frac{k_B T}{6\pi\eta R} \tag{1}$$

where D_{SE} is the diffusion coefficient of the particle, k_B is the Boltzmann constant, T is temperature, η is the viscosity of the matrix, and R is the radius of the particle. The Stokes–Einstein equation has been successfully applied to nanoparticles in chemically neutral (athermal) PNCs in the dilute NP loading regime where in the interparticle spacing (ID) is greater than the nanoparticle radius ($ID > R_{NP}$).[29−31] Nanoparticle diffusion is more complex in the presence of favorable nanoparticle–polymer interactions (attractive PNCs) and when the viscosity is heterogeneous near the NPs.[32]

Nanoparticle diffusion in attractive PNCs is complicated by competing time scales between the adsorption time of a polymer on the NP and the relaxation times of the polymer matrix. When the polymer is strongly adsorbed, such that the desorption time is long relative to the polymer reptation time, a core–shell model dominates. This has been demonstrated in strongly attractive PNCs (e.g., silica nanoparticle in poly(2-vinylpyridine))[33] and the NP diffusion coefficient is well-described by the core–shell model:[21,34,35]

$$R_{eff} = R_{NP} + R_g \tag{2}$$

$$D_{core-shell} = \frac{k_B T}{6\pi\eta_{PNC} R_{eff}} \tag{3}$$

where R_{eff} is the effective radius of the NP, R_{NP} is the nanoparticle radius, R_g is the radius of gyration of the polymer and the nominal thickness of the adsorbed polymer layer, η_{PNC} is the viscosity of the polymer nanocomposite accounting for the NP concentration, and T is the temperature in Kelvin. When the desorption time is shorter than the reptation time, the NP diffusion coefficient has an additional term to describe vehicular motion.[36]

We have recently demonstrated that time-of-flight secondary ion mass spectrometry can measure polymer and nanoparticle diffusion coefficients in polymer melts.[33,37,38] We have shown that the core–shell and vehicular mechanisms of diffusion exist in PNCs, and in the case of the vehicular mechanism, we were able to extract a desorption time. Now, we turn our attention to polymer-grafted nanoparticles, specifically large SiO_2 nanoparticles grafted with bottlebrush polymers. Previously, these PGNs have shown extended chain conformations when suspended in a good solvent for the bottlebrush grafted polymers and collapsed chain conformations in the neat state.[39] To our knowledge, there is not yet a good model for bottlebrush thickness as a function of grafting density and bottlebrush molecular weight in a polymer melt. By measuring the diffusion coefficient of the bottlebrush PGNs in a polymer matrix and applying the core–shell model, we will extract the bottlebrush thickness.

EXPERIMENTAL METHODS

Materials

Silica particles were synthesized by the Stöber method with an average radius of 80 ± 5.5 nm.[40] Bottlebrushes with backbones of poly(norbornene) and side chains of polystyrene were attached using a modified SI-ROMP method established previously (Figure 1).[39] The number-averaged molecular weight of the bottlebrushes (M_{BB}) was controlled by varying the degree of polymerization of the backbone with a fixed number-averaged side chain molecular weight (4.2 kDa, Đ = 1.02). These polymer-grafted nanoparticles (PGNs) were diffused into linear polystyrene (PS) standards of number-averaged molecular weights 100 kDa (Đ = 1.04) and 425 kDa (Đ = 1.10) that were purchased from Polymer Source. PGNs are labeled as PS-X, where X is M_{BB} (Table 1). Graft density of the bottlebrush polymers was determined via thermogravimetric analysis.[39]

Table 1. Characteristics of the Bottlebrush Polymer Grafted Nanoparticles, Specifically the Number-Averaged Molecular Weight and Dispersity of the Bottlebrush Polymer and the Graft Density

Polymer Grafted Nanoparticle (PGN)	M_{BB} (kDa)	Đ	Graft Density σ (# Chains/nm^2)
PS-80k	79.7	1.075	0.051
PS-164k	163.6	1.10	0.031
PS-274k	273.9	1.143	0.024
PS-388k	388.4	1.19	0.022

Dynamic Light Scattering (DLS)

Bottlebrush PGN solutions were prepared with 1 mg/mL PS-X in toluene and filtered through Titan3 1 μm PTFE. Dynamic light scattering was performed on a LS Spectrometer by LS Instruments with a Cobalt 300 mW 600 nm laser. The autocorrelation function was fitted to signal intensity at 25 °C for three measurements at 10 scattering angles evenly spaced from 30° to 130°. CONTIN analysis on this autocorrelation function resulted in a diffusion coefficient from which we compute the hydrodynamic radii of the bottlebrush PGNs in toluene. The results of all angles were averaged and used for data analysis.

Trilayer Sample Fabrication

Bottlebrush PGN diffusion was measured in trilayer samples comprised of a thin polymer nanocomposite between two thick homopolymer layers. Bottlebrush PGNs were suspended in a 50 g/L toluene solution in which the swollen bottlebrushes promote good dispersion. Separately, the polystyrene solutions (300 g/L) were prepared in toluene. Solutions were stirred for 24 h at 25 °C to ensure full dissolution. Following previously established methods,[37] these solutions are used to fabricate trilayer samples with two thick (~6 μm) PS films on either side of a thin (~400 nm) polymer nanocomposite (PNC) film. The base PS layer was created by spin coating 75 mL (~1500 rpm) of a PS solution onto a grown SiO_2 wafer for 1 min. The PNC midlayer was deposited onto a glass slide by spin coating (~1900 rpm, 1 min) 75 mL of a 5 vol% bottlebrush PGN solution made from mixing a PS-X solution and a PS solution. The top-layer was prepared by spin coating (~1500 rpm, 1 min) 75 mL of a PS solution onto a glass slide. Film thicknesses were estimated using scanning electron microscopy (SEM) of a trilayer cross-section referenced with spun coat predictions.[41] The trilayer sample was assembled by floating the PNC and top PS layer onto water and transferring them sequentially onto the base PS layer and drying at 50 °C. Trilayer samples were annealed at 170 °C under vacuum for 3−7 days. Cross sections of the trilayers were prepared by fracturing samples along a crystallographic plane of the silicon wafer. The backs of these specimens were coated with carbon paint suspended in IPA to reduce surface charging and increase signal. Additionally, the samples were sputter-coated in iridium at a 20 mA current, to a thickness of 4 nm to increase secondary ion yield.

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Macromolecules 2026, 59, 871−878

Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS)

ToF-SIMS measurements were performed in a TESCAN S8252X dual plasma FIB-SEM using 30-keV, 100 − 500 pA Xe^+ beam, and positive ion mode. Data was collected with 512 × 512 or 1024 × 1024 pixel resolution and a field of view (FoV) of 20 × 20 μm^2 for 400 frames. Variations in current and pixel count are specified in Supporting Information. ToF SIMS analysis was performed on a 3D data set from the trilayer film and used to track the signal for m/Q = 28, which corresponds to Si^+ from the SiO_2 nanoparticles (Figure 2). 1D concentration profiles were extracted from the 3D intensity data by correcting for sample orientation, integrating perpendicular to the plane of highest nanoparticle concentration, and setting the center of the PNC layer to $x = 0$.[37] This ion intensity profile was iteratively fit to Fick's second law for a finite source diffusing into a semi-infinite medium using

$$\varphi(x) = \frac{1}{2}\left[\mathrm{erf}\left(\frac{h-x}{\sqrt{4D_{NP}t}}\right) + \mathrm{erf}\left(\frac{h+x}{\sqrt{4D_{NP}t}}\right)\right] \quad (4)$$

where φ is the volume fraction of nanoparticles, h is the initial PNC film thickness in μm, x is the distance of diffusion in μm, t is the diffusion time in seconds, and D_{NP} is the diffusion coefficient of the bottlebrush PGN in cm^2/s. Diffusion coefficients are measured at multiple annealing times to demonstrate time independence.[37] The 1D concentration profiles for all samples are provided in Figures S2−S6.

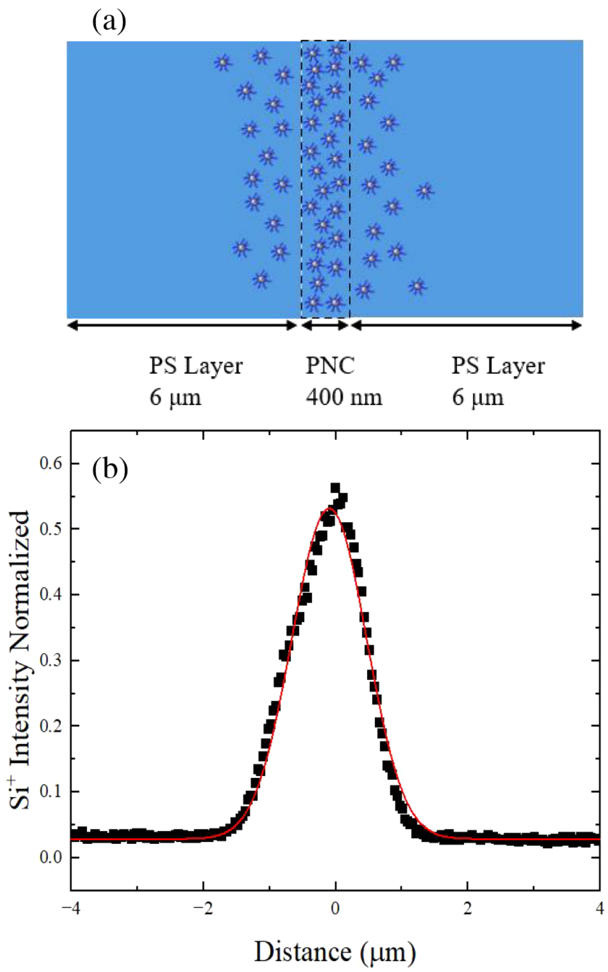

Figure 2. (a) Schematic of PS/PNC/PS trilayer sample, after annealing. (b) A tilt-corrected intensity profile for m/Q = 28 corresponding to Si from the nanoparticles. Sample annealed at 170 °C for 5 days. eq 4 finds $D_{NP} = 3.01 \times 10^{-15}$ cm^2/s.

■ RESULTS AND DISCUSSION

Diffusion of Bottlebrush Polymer-Grafted Nanoparticles

The bottlebrush polymers grafted on large SiO_2 nanoparticles have bottlebrush molecular weights (M_{BB}) from 80 to 388 kDa (Table 1). Each monomeric unit in the bottlebrush backbone has 40:1 styrene to norbornene monomer units, and thus the grafted bottlebrush is effectively polystyrene. To study how the M_{BB} impacts the effective radius of the nanoparticle in a dilute solution, the bottlebrush PGNs were placed in toluene, a good solvent for polystyrene. As expected, the bottlebrush thickness (t_{BB}) in these PGNs increases with M_{BB} (Figure 3). The bottlebrush thicknesses were previously measured by TEM in the absence of solvent,[39] and t_{BB} is significantly larger in a good solvent, as expected. Moreover, the bottlebrush thicknesses increase only slightly with M_{BB} in the neat state, while t_{BB} increases significantly when swollen by a good solvent.

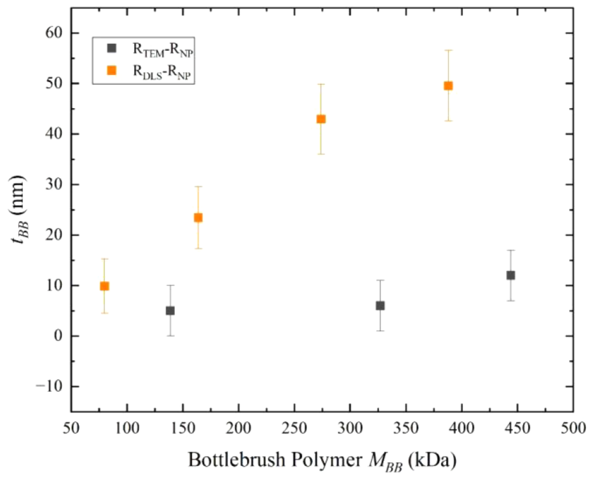

Figure 3. Bottlebrush thickness (t_{BB}) characterized by DLS (solution) and TEM (neat).

The diffusion coefficients of bottlebrush PGNs, D_{NP}, in PS matrices of molecular weights 100 kDa and 425 kDa were measured using ToF-SIMS (Figures S2−S6). In the unannealed trilayer samples, the bottlebrush PGN volume fraction is 5%, which is sufficient for this PNC to be considered dilute. The bottlebrush PGN diffusion coefficients in 100 kDa linear polystyrene are plotted with respect to M_{BB} (Figure 4). There is sufficient agreement between the annealing times to demonstrate time-independent diffusion. Overall, larger grafted bottlebrushes correspond to slower bottlebrush PGN diffusion. This is attributed to the grafted bottlebrush layer thickness increasing with increasing M_{BB}, which is qualitatively consistent with Figure 3.

In the core−shell model of NP diffusion, the grafted bottlebrush polymer is the shell that covers the open sites of the silica surface. The effective radius of bottlebrush PGNs, $R_{eff\text{-}BB}$, is defined by

$$R_{eff-BB} = R_{NP} + t_{BB} \quad (5)$$

where R_{NP} is the silica nanoparticle radius and t_{BB} is the thickness of the grafted bottlebrush polymer in the PS matrix. In this system, a reasonable range of t_{BB} is 10 to 30 nm, as estimated from the measured t_{BB} values in the neat and

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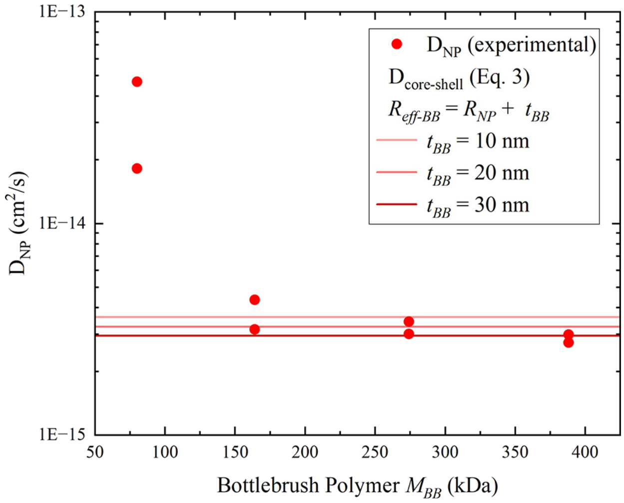

Figure 4. Diffusion coefficients of PS-X nanoparticles plotted against the molecular weight of the bottlebrush chains at 170 °C in 100 kDa PS matrix. The core−shell model is shown for three values of t_{BB} across the expected range of PGN grafted layer thicknesses (10−30 nm), as shown by solid red lines.

solution state (Figure 3). Note that t_{BB} in a polymer nanocomposite is expected to be larger than in the PGN-only sample measured by TEM due to the penetration of the matrix polymer into the grafted polymer to swell the bottlebrush. This establishes the lower bound of possible t_{BB} in the melt, while the solution state with extended bottlebrushes establishes the upper bound. Bottlebrush PGN diffusion coefficients were computed using the core−shell model (eq 3), polystyrene viscosity (100 kDa, 170 °C, 1×10^4 Pa·s), and three values of t_{BB} (10, 20, and 30 nm) were chosen to represent viable range of t_{BB} between the neat and solution state bounds, Figure 4 (lines). Only when the bottlebrush molecular weight is >150 kDa do the computed diffusion coefficients agree well with the measured diffusion coefficients for these bottlebrush PGNs. More specifically, the quality of the agreement is best when t_{BB} = 10 nm for PS-164k, t_{BB} = 20 nm for PS-274k, and t_{BB} = 30 nm for PS-388k. This finding is consistent with an increase $R_{eff\text{-}BB}$ associated with increasing bottlebrush molecular weight and slower bottlebrush PGN diffusion according to the core−shell model. Notably, the diffusion coefficient of the PS-80k PGNs is an order of magnitude higher than predicted by the core−shell model, which suggests that the viscosity near the PGNs is unexpectedly low when the bottlebrush molecular weight is small (<125 kDa).

Local Entanglement Density Effects on Bottlebrush PGN Diffusion

The interpenetration between grafted and matrix polymers that are chemically similar is a balance of entropically favorable mixing and entropically unfavorable stretching of the grafted polymer.[42−45] When the molecular weight of the matrix polymer is high relative to the grafted polymer, the grafted polymers collapse and dewet from the surrounding polymer matrix.[47−50] This phenomenon increases the degree of confinement of the grafted polymers and reduces the interpenetration between the grafted and matrix polymers.[46] Consequently, the matrix polymers adjacent to the grafted polymers are less entangled, as previously described in polymers adjacent to impenetrable surfaces.[50−54] When the local entanglement density is lower, the polymer melt viscosity (η) is lower and the molecular weight dependence is weaker ($\eta \propto M^a$) with the exponent a dropping from 3.4 in well-entangled melts to 1.0, as predicted by the Rouse theory of unentangled melts.[55,56] Thus, when a matrix polymer dewets from a PGN, the local decrease in entanglement density produces a local decrease in melt viscosity.

To explore the impact of the matrix molecular weight on the local entanglement density, we estimate the local matrix viscosity using the core−shell model (eq 3), defining the calculated value as $\eta_{core-shell}$. For this calculation, we use the measured D_{NP} and the following bottlebrush polymer thicknesses: 10 nm for PS-80k and PS-164k, 20 nm for PS-274k, and 30 nm for PS-388k. We plot the calculated viscosities, $\eta_{core-shell}$, normalized by the bulk viscosity of neat PS at 170 °C ($\eta_{PS} = 1 \times 10^4$ Pa·s for 100 kDa; $\eta_{PS} = 1.5 \times 10^7$ Pa·s for 425 kDa) versus the bottlebrush molecular weight (M_{BB}) normalized by the molecular weight of the PS matrix (M_{PS}) (Figure 5). When the bottlebrush molecular weight is

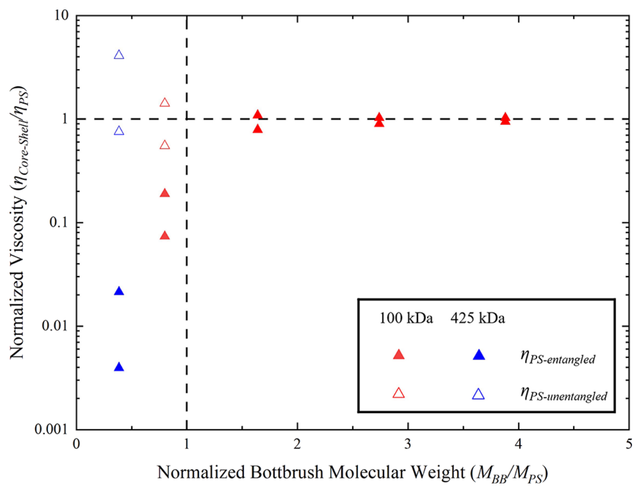

Figure 5. Computed matrix viscosity from the core−shell model (eq 3) normalized by the PS homopolymer viscosity versus the bottlebrush molecular weight M_{BB} normalized by the PS matrix molecular weight, M_{PS}: 100 kDa (red) or 425 kDa (blue). The PS homopolymer viscosity was calculated using entangled PS viscosities (closed triangles) and extrapolated unentangled PS viscosities (open triangles).

greater than the matrix molecular weight, the normalized viscosity is unity, indicating that $\eta_{core-shell} = \eta_{PS}$, and demonstrating that the local entanglement density is unperturbed by the nanoparticles. Similar to Figure 4, applying t_{BB} values that increase with M_{BB} yields excellent agreement with the core−shell model when the matrix polymer is >150 kDa, corresponding to $M_{BB}/M_{PS} > 1$. When the bottlebrush molecular weight is smaller than the matrix molecular weight, the melt viscosity from the core−shell model is 10−100 times smaller than the entangled PS viscosity (filled symbols): $\eta_{core-shell} \ll \eta_{PS}$. This was observed for PS matrix molecular weights of 100 kDa (red) and 425 kDa (blue), and the decrease in $\eta_{core-shell}/\eta_{PS}$ is more pronounced for smaller M_{BB}/M_{PS}. These results suggest that short, grafted bottlebrush polymers dewet from the matrix polymer, which subsequently reduces both the local entanglement density and local viscosity in the matrix polymer. Lower local viscosity near bottlebrush PGNs with low bottlebrush molecular weights is consistent

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Macromolecules 2026, 59, 871−878

with the unexpectedly fast diffusion of PS-80k in 100 kDa PS in Figure 4.

Next, we estimate the viscosity of the PS matrix assuming unentangled polymer conformations by extrapolating the unentangled PS behavior at low molecular weights to higher molecular weights at 170 °C. We extrapolate values from literature to estimate an unentangled $\eta_{PS} = 1.33 \times 10^3$ Pa·s for 100 kDa and $\eta_{PS} = 8.02 \times 10^3$ Pa·s for 425 kDa (eq. S4, Figure S9). When $M_{BB}/M_{PS} < 1$, normalizing the $\eta_{core-shell}$ values computed from the diffusion coefficients by these unentangled viscosities yields values close to one ($\eta_{core-shell} \approx \eta_{PS}$), Figure 5 (open triangles). This finding demonstrates that the relevant viscosity for NP diffusion when $M_{BB}/M_{PS} < 1$ is given by the unentangled PS viscosities. Thus, when $M_{BB} < M_{PS}$ and the bottlebrush polymer dewets from the matrix polymer, the local decrease in entanglement density reduces the local viscosity and accelerates NP diffusion. Moreover, this reduced viscosity is comparable to that of an unentangled polystyrene matrix of the same molecular weight.

Across the M_{BB}/M_{PS} range studied with these bottlebrush PGNs, the nanoparticle diffusion in a polymer is governed by the core−shell model, which is the Stokes−Einstein relationship accounting for changes in the nanoparticle size due to the grafted polymer and the matrix viscosity associated with autophobic dewetting. The matrix viscosity is lower near the grafted polymer when the bottlebrush is shorter (Figure 6a) and uniform when the bottlebrush is longer (Figure 6b). The schematics in Figure 6 illustrate the nanoparticle surfaces as flat to highlight the large size of the nanoparticles (R_{NP} = 80 nm) relative to the grafted bottlebrush thicknesses. The propensity for dewetting is also driven by the higher grafting density for lower molecular weight bottlebrush polymer: σ(PS-80k) = 0.051 chains/nm^2 and σ(PS-388k) = 0.022 chains/nm^2 (Table 1). Dewetted PGNs form a dry bound layer at the nanoparticle surface because the matrix polymer is unable to penetrate the dense bottlebrush, which reduces the local entanglement density and viscosity. Far away from the dewetted bottlebrush PGN, the polymer matrix is expected to regain the bulk entanglement density and viscosity.

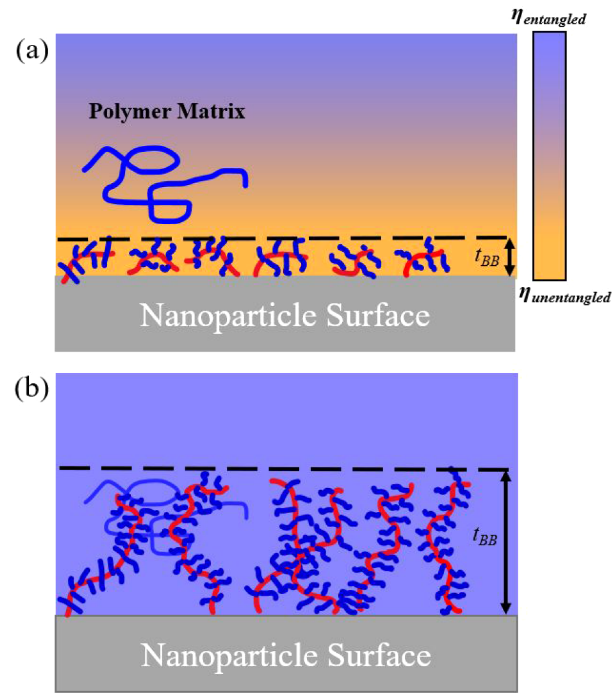

Figure 6. (a) Schematic of a dry PGN surface when $M_{BB} < M_{PS}$ with a dewet bottlebrush polymer height of t_{BB}. The background gradient color qualitatively indicates the reduced matrix viscosity near the PGN surface. (b) Schematic of a wet PGN surface when $M_{BB} > M_{PS}$ with a swollen bottlebrush polymer and uniform matrix viscosity.

The diffusion of these nanoparticles with short, grafted bottlebrushes is distinct from nanoparticles with adsorbed polymer. Our recent studies of nanoparticles diffusing in polymer melts have considered neutral and attractive NP−polymer interactions.[2,3,30,33,37,38] When the interactions are neutral and there is no adsorbed polymer, the NPs diffuse according to the core−shell model with the bulk matrix viscosity and $R_{eff} = R_{NP}$.[30,37] When the interactions are attractive and long-lived relative to the time scale of diffusion, the NP is slower as described by the core−shell model with $R_{eff} = R_{NP} + R_g$, where R_g is the radius of gyration of the matrix polymer and the nominal thickness of the adsorbed polymer layer on the NP.[30,33] The polymers adsorbed on the NPs incorporate matrix polymers, and the polymer conformations are perturbed primarily within R_g from the nanoparticle.[2,3,38] Consequently, nanoparticle diffusion in systems with attractive NP−polymer interactions is well-described by the core−shell model when using the bulk viscosity. While a bound layer is present in these systems, it does not exclude the surrounding matrix and make the dry layer required for enhanced diffusion. These earlier findings are in marked contrast to this report of dry polymer brushes disrupting the polymer conformations of the neighboring matrix polymers, reducing the local viscosity, and increasing the NP diffusion coefficient when $M_{BB}/M_{PS} < 1$.

The nature of polymer-grafted nanoparticle diffusion also depends on the architecture of the grafted polymer. A previous study investigated an athermal system comprised of linear PGNs diffusing in homopolymers.[21] The diffusion of nanoparticles grafted with linear polymers (16−21 kDa) was well-described by the core−shell model when the thickness of the shell is the nominal thickness of the grafted layer wetted by the matrix polymer.[21] This finding was consistent across a range of the matrix molecular weights (4−51 kDa) and grafting densities (0.17−0.55 chains/nm^2). These linear PGNs were wet by the matrix polymer across a range of grafting densities and matrix molecular weights, which is in contrast to bottlebrush PGNs studied here, in which autophobic dewetting was found at $M_{BB}/M_{PS} < 1$. Enhanced diffusion was not observed in linear PGNs as no dewetting occurred for the surrounding matrix was not of high enough molecular weight to make interpenetration entropically unfavorable. Overall, our study highlights that there is more to understand about the conformations and wetting of polymers grafted to nanoparticles, particularly as a function of grafted polymer architecture, grafting densities, and molecular weights, and nanoparticle diffusion can provide valuable insights into these complex systems.

■ CONCLUSIONS

We measured diffusion coefficients of nanoparticles grafted with bottlebrush polymers in linear homopolymers with athermal interactions and showed excellent agreement with the core−shell model. When the bottlebrush molecular weight is greater than the matrix polymer molecular weight and the brush is wet, the core−shell model fits the data using the bulk viscosity. This model was used to extract the thickness of the

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Macromolecules 2026, 59, 871−878

bottlebrush layers, which was found to increase with bottlebrush molecular weight as expected. When the bottlebrush grafted polymer dewets from the matrix, the nanoparticle diffusion is unexpectedly fast. We attribute this finding to a local decrease in the matrix viscosity and found quantitative agreement when we approximated the reduced viscosity by extrapolating from the unentangled viscosities to higher molecular weights corresponding to the polymer matrix. We demonstrate that by measuring the NP diffusion coefficients, we can differentiate between the wet and dry states of a grafted bottlebrush polymer, which has been previously difficult to discern for nanoparticles with nonlinear grafter polymers. Additional characteristics of nanoparticles with grafted polymers in polymer matrices are yet to be experimentally explored to further understand both nanoparticle diffusion and local polymer conformations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.5c02957.

> ToF-SIMS mass spectra distinguished conditions m/Q = 28 (Si+) peak; table of measured diffusion coefficients for PGNS in PS (100 kDa and 425 kDa) matrices; normalized concentration profiles for PS-80k, PS-164k, PS-274k, and PS-388k PGNs in 100 kDa PS matrix; normalized concentration profiles for PS-164k PGN in 425 kDa PS matrix; hydrodynamic radius as a function of scattering angle of PS-X samples in toluene; and estimate of the unentangled melt viscosity for bulk PS (PDF)

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Notes

The authors declare no competing financial interest.

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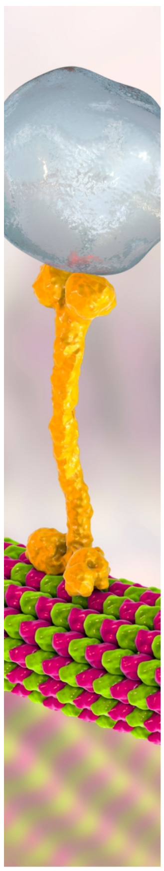

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